

Science Tools & Safety

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Match the tool to the job: a **hand lens** to look closely, a **balance** for mass, a **graduated cylinder** for liquid volume, a **thermometer** for temperature, and a **ruler** for length. Safety first: goggles on, walk, follow directions, and clean up.

Round 1 — Retrieve It

1. Which tool would you use to measure the mass of a rock?

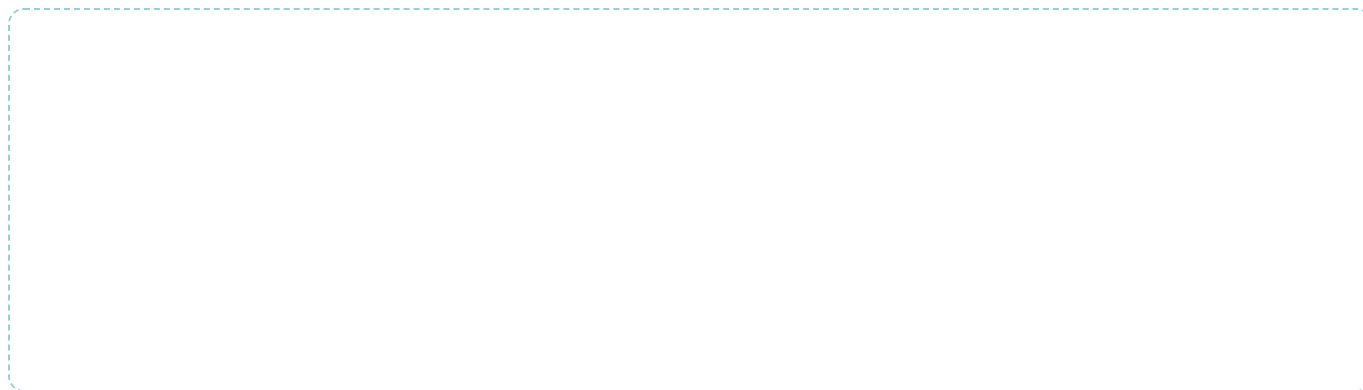
2. Which tool measures how much liquid is in a container?

3. Which tool helps you see the tiny parts of a leaf?

4. Name one thing you should always do to stay safe in a science lab.

Round 2 — Sketch It

Draw a graduated cylinder with water in it. Show where your eyes should be to read the amount correctly (at eye level with the curve of the water).



Science Tools & Safety — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Tool

1. You want to find out which of two toy cars is longer. Which tool and unit would you use?

2. A recipe needs 50 mL of water. Which tool measures that, and how do you read it?

3. You want to know if it is warmer inside or outside. Which tool tells you, and what unit does it use?

Round 4 — Reason from Evidence

A student pours water into a graduated cylinder and looks down at it from above. Their reading is different from their partner's, who looks at eye level. Whose reading should the class trust, and why?

Science Tools & Safety — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why scientists use tools instead of just guessing with their senses. Give one example.

C**Connect**

Connect a safety rule to what could go wrong without it. Pick one rule and describe the problem it prevents.

E**Extend**

Design a quick investigation and list every tool you would need, with the unit each one measures.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Science Tools & Safety

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A balance (measures mass in grams).
2. A graduated cylinder (measures liquid volume in mL).
3. A hand lens (magnifier).
4. Any: wear goggles, walk don't run, follow directions, tie back hair, clean up, tell the teacher about spills.

Sketch It — look for

Eyes shown at the level of the water's curved surface (the meniscus), reading the bottom of the curve.

Apply It

1. A ruler (or meter stick), measured in centimeters (cm).
2. A graduated cylinder; read at eye level at the bottom of the water's curve.
3. A thermometer, measured in degrees Celsius (°C).

Reason from Evidence — look for

Trust the partner reading at eye level. Looking down distorts the measurement; a graduated cylinder must be read straight on at the meniscus for an accurate volume.

ACE — Extend look-for

Tools listed match the investigation and each is paired with the correct unit (mass → g, volume → mL, length → cm, temperature → °C).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.